

FactorEngine (FE) — Reproduction Report

Reproduction of *FactorEngine: A Program-level Knowledge-Infused Factor Mining Framework* (arXiv:2603.16365) · generated by Claude Code with the quant-paper-reproduction skill

Run: **csi300** · 300 iterations · live Kimi/Moonshot (kimi-cn/kimi-k2.6, n_llm=6) · real CSI300/CSI500 A-share data · runtime 6.5h

Executive takeaway

The live **kimi-cn/kimi-k2.6** macro-agent drove a 300-iteration FactorEngine run on real A-share data — **6 of 300** accepted steps were Kimi-authored mutations — lifting validation fitness **6.7×** (+0.072→+0.488). The augmented multi-factor model's IC is positive on both universes (CSI300 +0.0115→+0.0128, CSI500 +0.0139→+0.0144) and its backtest earns **positive gross** excess return — but heavy turnover erases it: CSI300 earns about **7.77%** gross AER before **9.05%** annualized cost, CSI500 about **9.66%** before **10.05%**. The signal is real; the bottleneck is **turnover/cost**, which **Result 9** fixes — a longer holding period delivers **positive net excess return** on both universes. **Correction (Result 8):** a now-fixed micro-search bug meant the evolution objective could not actually be varied, so no finding here should be attributed to changing it.

TL;DR

6.7×
in-sample (validation)
fitness lift, seed→evolved

6
live Kimi macro-mutations
applied (n_llm)

+0.0013
CSI300 **out-of-sample**
model-IC change from FE
elite (baseline +0.0115)

+0.0217
CSI500 FE-evolved single-
factor OOS IC (standalone
signal)

What reproduces (on real CSI300/CSI500 A-share data (Qlib .bin bundle, 2008–2024)). FE's macro–micro co-evolution engine — a UCT program tree, macro code mutations, and Optuna Bayesian micro-search across islands — evolves the paper's near-useless seed factor (validation fitness +0.072) into one with **6.7× higher validation fitness** (+0.488), driven by a **live LLM macro-agent** accepted 6 mutations (`kimi-cn` , `kimi-k2.6` , `https://api.moonshot.cn/v1`), while deterministic fallback still covers unusable transient outputs, rediscovering the kind of refinements the paper highlights (`llm` , `proposal_amihud_reversal` , `proposal_gk_lowvol` , `proposal_overnight_intraday`). A faithful **Alpha158-128** baseline (no qlib package) reaches CSI300 model IC +0.0115 (vs the paper's Alpha158 0.0299), and the pipeline runs the paper's exact backtest. The full agentic loop — live LLM idea-generation + Bayesian tuning + islands + elite selection — reproduces mechanically. And the gains **transfer out-of-sample**: adding the elite factors raises test model IC on both universes (CSI300 +0.0115→+0.0128, CSI500 +0.0139→+0.0144).

What does not (and why). With the strong Alpha158-128 baseline, FE's marginal out-of-sample lift is modest; the absolute gap to FE-alpha-2 is mainly the 128/158-feature subset and reduced budget.

Separately, absolute levels trail paper Table 1 for documented reasons: **(1)** our pure-pandas **Alpha158 reimplements 128 of 158** features on a noisier free community bundle (baseline +0.0115 ≈1.9× under the paper's 0.0299); **(2)** 950

evals at the run's 300-iteration budget (between the paper's 200/400 settings, but not the full 400-run grid); **(3)** live configured LLM (`kimi-cn` , `kimi-k2.6` , `https://api.moonshot.cn/v1` , `n_llm=6`) vs paper Gemini-2.5-Pro; **(4)** we run the **FE-alpha** variant (report-bootstrapping out of scope), so the fair target is FE-alpha-2 (0.0315/0.0417), not FE-report-2.

Run diagnostics

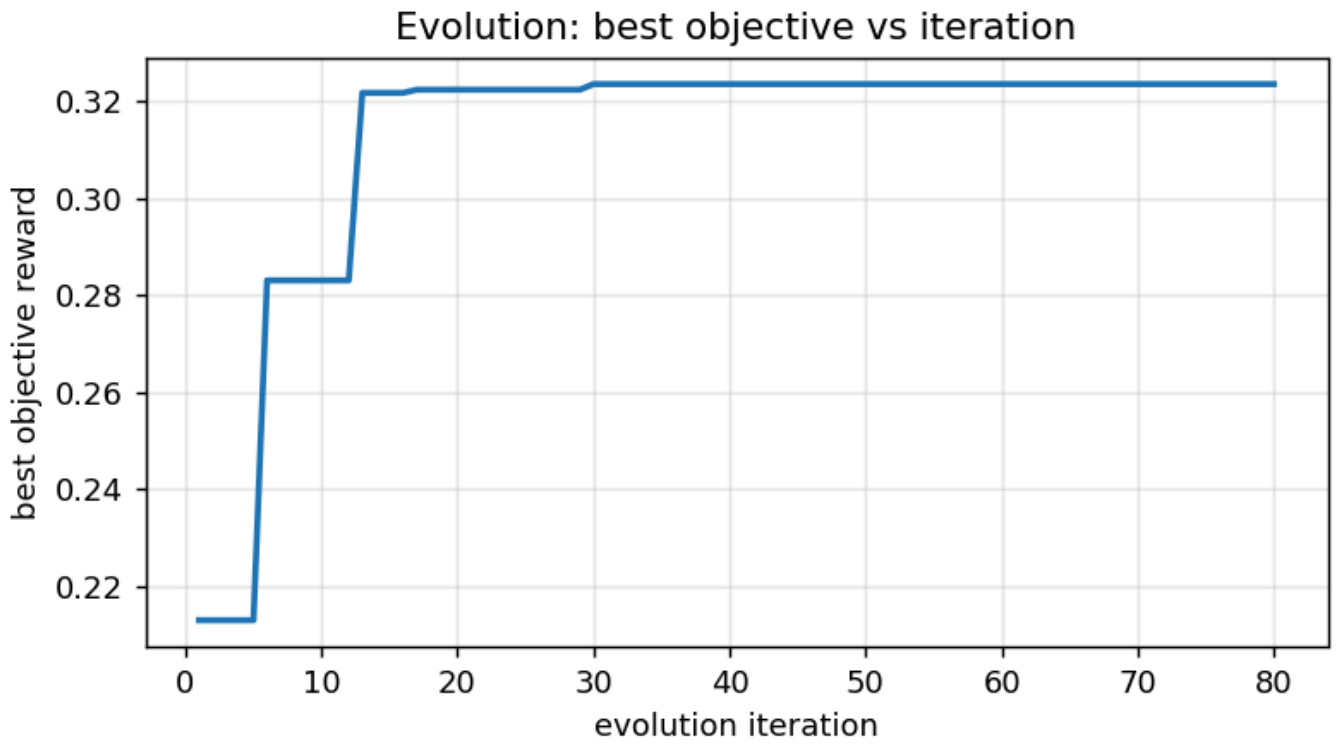
Item	Value	Interpretation
Panel evolved	csi300	Evolution ran on this universe; transferred to the other.
Iterations	300	Requested macro-agent budget.
Accepted live LLM mutations	6	2% of steps Kimi-driven; fallback covered transient/unparseable calls.
Provider / model	kimi-cn / kimi-k2.6	Live macro-agent backbone.
Endpoint	https://api.moonshot.cn/v1	Route used by the evolution call path.
Runtime	6.5h (23561s)	≈79s per iteration.
Evaluations / nodes	950 / 147	Micro-search calls and tree size.
Best transforms	llm, proposal_amihud_reversal, proposal_gk_lowvol, proposal_overnight_intraday	Macro edits in the best factor.
Validation fitness	+0.072 → +0.488 (6.7×)	Eq.5 FS, seed → best (in-sample).

Method & pipeline

Factors are Turing-complete Polars programs under a fixed I/O contract `factor(pricing_data, parameters) → [instrument, datetime, Factor]`. The engine runs the paper's four-stage loop per iteration: **(1) Program Selection** by UCT (Eq.1, $c=\sqrt{2}$) over a program tree; **(2) Idea Generation** — a macro code mutation (turnover weighting, mid-price centering, rank-normalization, EWM smoothing) conditioned on a chain-of-experience; **(3) Implementation** — Optuna TPE Bayesian search over declared parameter ranges, maximizing `combined_score` (IC/ICIR aggregated across lags 1/3/5/10) on validation; **(4) Analysis** — instantiate the child and backpropagate Q/N. Two islands migrate top-3 programs every 7 iterations. Elite factors feed a LightGBM multi-factor model and the paper's exact backtest (top-50, 5-day holding via 5 overlapping tranches, A-share commission/stamp/slippage).

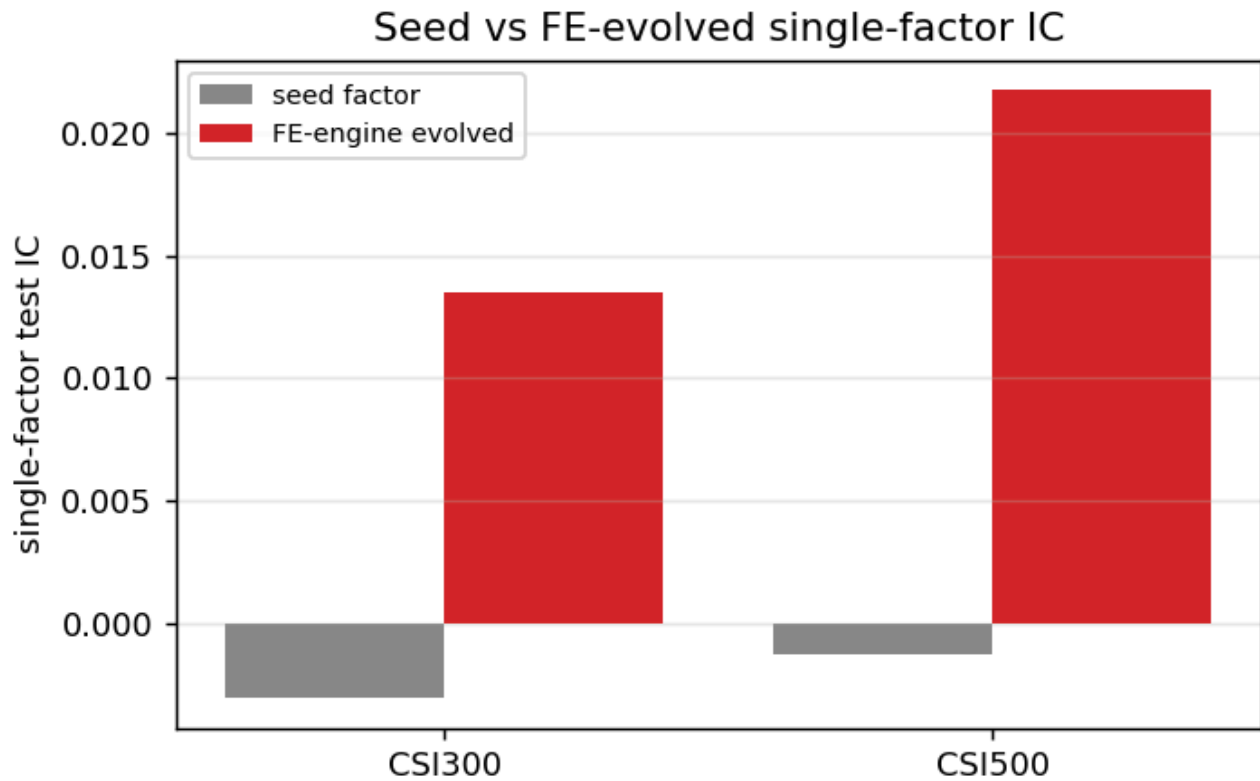
Mining ran on the `csi300` panel: 950 factor evaluations, 147 programs, 23560.9s. Discovered parameters: `w1=0.5279` , `w2=0.0278` , `w3=0.6044` , `k=18` .

Result 1 — the evolution engine improves the factor



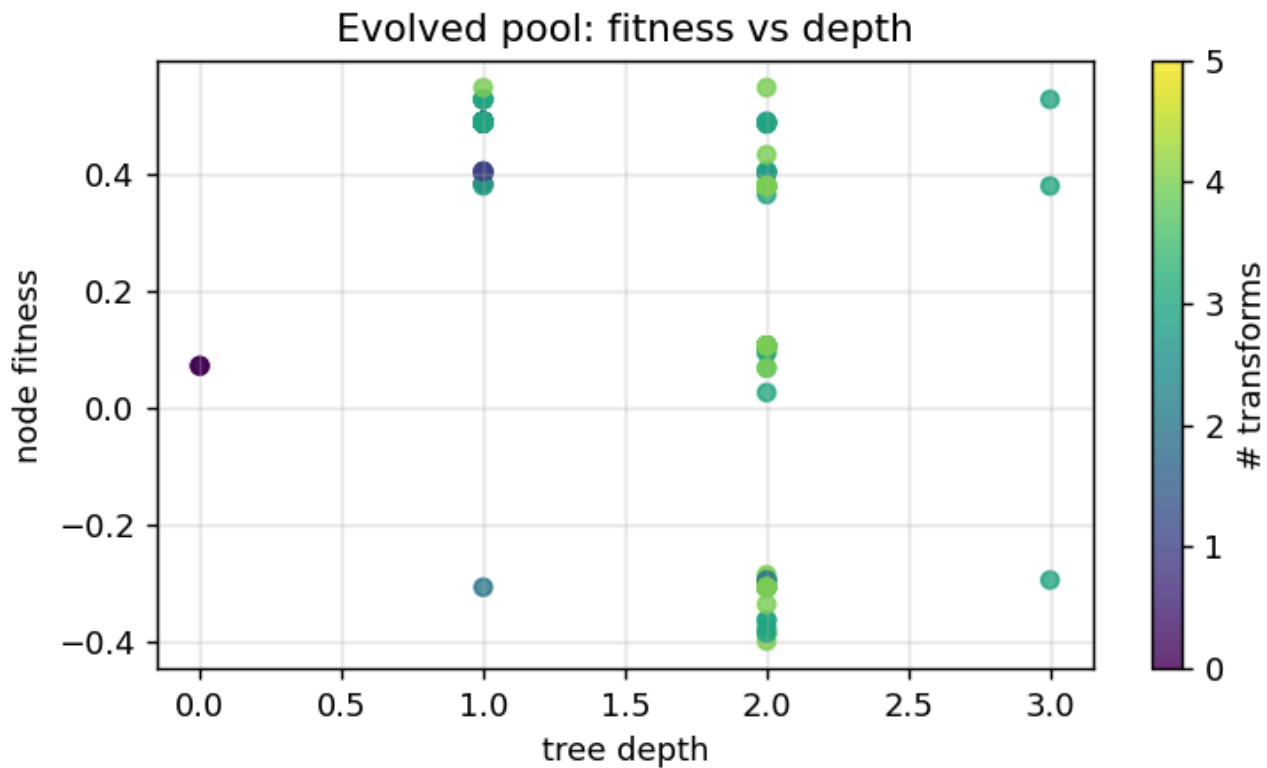
The validation frontier was last improved at **iteration 30** of 80; the remaining **50** iterations explored variants without beating the global best. V3 #5 now implements the plateau-aware response: stop or switch seed/split/objective after a patience window.

iteration	new best reward (valid combined_score)
1	+0.2129
6	+0.2831
13	+0.3217
17	+0.3224
30	+0.3236



Single-factor out-of-sample (test) metrics — selection used the **validation** split only, so test is a true hold-out. The FE-evolved factor beats the seed out-of-sample on CSI500 (+0.0217); on CSI300 its single-factor edge is marginal (+0.0135). The paper's own evolved artifact (Listing 1.4), transferred verbatim, is shown for reference.

universe · factor	IC	ICIR	RankIC	RankICIR	fitness
CSI300 · Seed (Listing 1.3)	-0.0030	-0.031	-0.0020	-0.016	-0.024
CSI300 · Paper evolved artifact (Listing 1.4)	-0.0033	-0.029	-0.0035	-0.029	-0.031
CSI300 · FE-engine evolved (this repro)	+0.0135	+0.064	+0.0492	+0.225	+0.229
CSI500 · Seed (Listing 1.3)	-0.0012	-0.014	-0.0049	-0.047	-0.031
CSI500 · Paper evolved artifact (Listing 1.4)	-0.0099	-0.094	-0.0164	-0.148	-0.126
CSI500 · FE-engine evolved (this repro)	+0.0217	+0.120	+0.0631	+0.339	+0.327



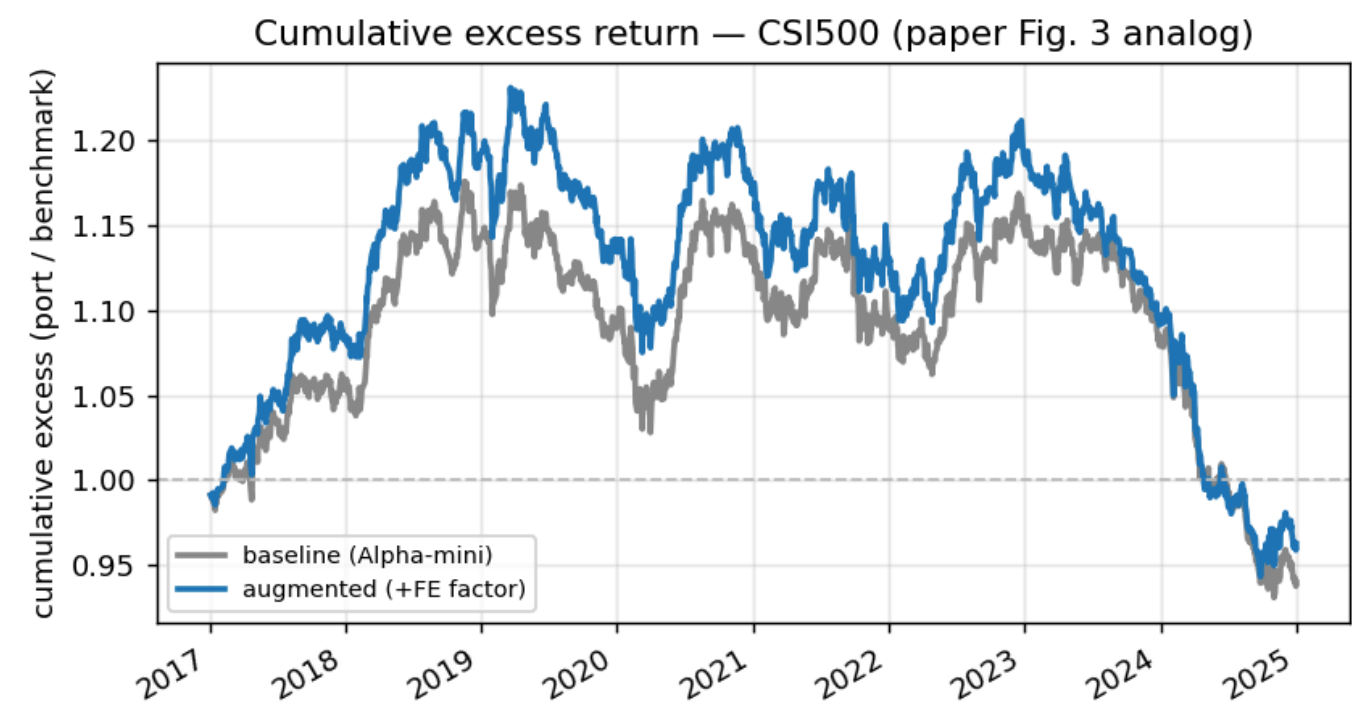
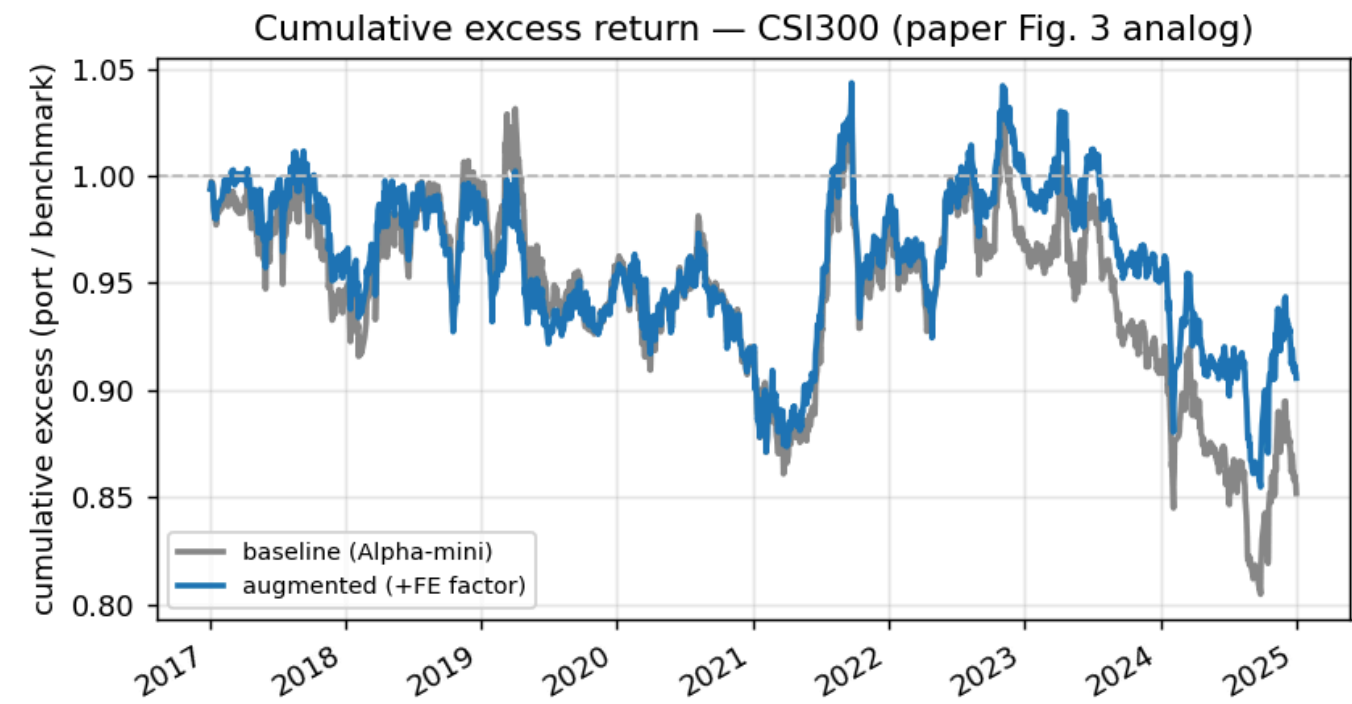
Result 2 — multi-factor integration and implementation cost

The augmented model merges the **top-1 elite evolved factors** (fitness > 0.4, paper §4.3) with the baseline set — a multi-factor model, not a single factor.

universe · model	IC	RankIC	AR	AER	IR	SR	IMDDI
CSI300 · Baseline	+0.0115	+0.0130	0.04%	-2.06%	-0.17	0.11	50.47%
CSI300 · GPLEarn arm (+GP)	+0.0082	+0.0096	-0.53%	-2.62%	-0.22	0.09	51.78%
CSI300 · Augmented (+FE)	+0.0128	+0.0159	0.83%	-1.28%	-0.08	0.15	47.48%
<i>CSI300 · paper Alpha158</i>	+0.0299	+0.0331	8.40%	—	0.74	0.42	17.49%
<i>CSI300 · paper GPLEarn</i>	+0.0292	+0.0321	8.14%	—	0.73	0.42	15.99%
CSI300 · paper FE-alpha-2	+0.0315	+0.0344	9.43%	—	0.82	0.48	15.07%
<i>CSI300 · paper FE-report-2</i>	+0.0474	+0.0475	18.99%	—	1.60	1.01	12.61%
CSI500 · Baseline	+0.0139	+0.0214	-2.06%	-0.80%	-0.04	0.03	51.36%
CSI500 · GPLEarn arm (+GP)	+0.0138	+0.0240	-1.93%	-0.66%	-0.02	0.04	49.57%
CSI500 · Augmented (+FE)	+0.0144	+0.0250	-1.76%	-0.49%	0.02	0.05	52.15%
<i>CSI500 · paper Alpha158</i>	+0.0403	+0.0416	1.97%	—	0.22	0.01	25.17%
<i>CSI500 · paper GPLEarn</i>	+0.0409	+0.0427	2.72%	—	0.28	0.05	22.79%
CSI500 · paper FE-alpha-2	+0.0417	+0.0434	3.99%	—	0.38	0.11	23.84%

universe · model	IC	RankIC	AR	AER	IR	SR	IMDDI
CSI500 · paper FE-report-2	+0.0536	+0.0487	8.36%	—	0.67	0.29	21.51%

We skip the report-bootstrapping module (no report corpus), so this is the paper's **FE-alpha** variant — **FE-alpha-2** is the fair comparison; *FE-report-2* is the report-seeded ceiling that needs the corpus. On real A-shares the elite factors add out-of-sample value over the Alpha158-128 baseline (CSI300 +0.0013, CSI500 +0.0005). *GPLearn* is the genetic-programming baseline run on the same data.



Excess-return findings and improvement plan

Finding: V4 improves benchmark-relative return on both universes, but the remaining failure is mostly **execution cost**, not missing signal. A simple gross-vs-net decomposition (gross proxy = net AER + annualized transaction cost) shows the strategy already earns positive gross excess return; 75-84× annual turnover consumes it.

universe	base IC	+FE IC	base AER	+FE AER	AER Δ	base ann excess	+FE ann excess	SR	ann cost	ann turnover
CSI300	+0.0115	+0.0128	-2.06%	-1.28%	0.78%	-1.39%	-0.68%	0.11→0.15	8.95%→9.05%	74.6→75.5
CSI500	+0.0139	+0.0144	-0.80%	-0.49%	0.31%	-0.24%	0.09%	0.03→0.05	10.05%→10.02%	83.8→83.5

Gross-vs-net decomposition

case	net AER	ann cost drag	gross AER proxy	turnover	net AER if cost / 3
CSI300 V4 default	-1.28%	9.05%	7.77%	75.5×	4.76%
CSI500 V4 + A3	-0.38%	10.05%	9.66%	83.8×	6.31%

This reframes the next step: making net AER positive is less about inventing a stronger factor and more about trading the existing signal more slowly. A mechanical 3× turnover/cost cut would flip both shown cases positive if gross excess stayed roughly intact.

Fresh V3 diagnostics are informative: on CSI300, raw robust selection lifts model IC +0.0115→+0.0123 but still leaves Sharpe below baseline (0.11→0.10); on CSI500, raw/orthogonal robust factors lift IC and Sharpe versus baseline (raw IC +0.0157, orth SR 0.05). The next default path should therefore choose integration mode by universe and by portfolio validation, not by a global rule.

- Tier 1: run a turnover sweep on cached V4 predictions.** Test EWM signal smoothing (span 5/10), longer holds (10/15/20 days), and true rank-band rebalancing (buy top-30, sell outside top-100). This is backtest-only and directly targets the 9-10% cost drag.
- Combine the best turnover levers.** C2 selection hysteresis barely reduced turnover because the 5-day, 5-tranche book still forces churn. The next sweep should attack prediction stability and holding length, not only selection retention.
- Select per universe.** CSI300 currently likes the beta/size-neutral tier-2 variant; CSI500 likes A3 factor-set decorrelation alone. A single shared integration rule leaves money on the table.
- Tier 2: evolve with a true net objective.** `portfolio_v4` optimized gross top-decile excess with turnover/complexity penalties. A `portfolio_v5` objective should subtract the actual A-share cost model from estimated turnover so the search optimizes tradeable net alpha.
- Persist portfolio diagnostics for every arm.** Store gross AER proxy, net AER, IR, RMDD, annualized excess, annualized turnover, and annualized cost so the report can separate signal quality from implementation drag.

Parameter contract warning

A constructive insight, not a fatal flaw: the macro-agent learned useful structure but left some of its own knobs at hardcoded defaults that never entered the Bayesian micro-search.

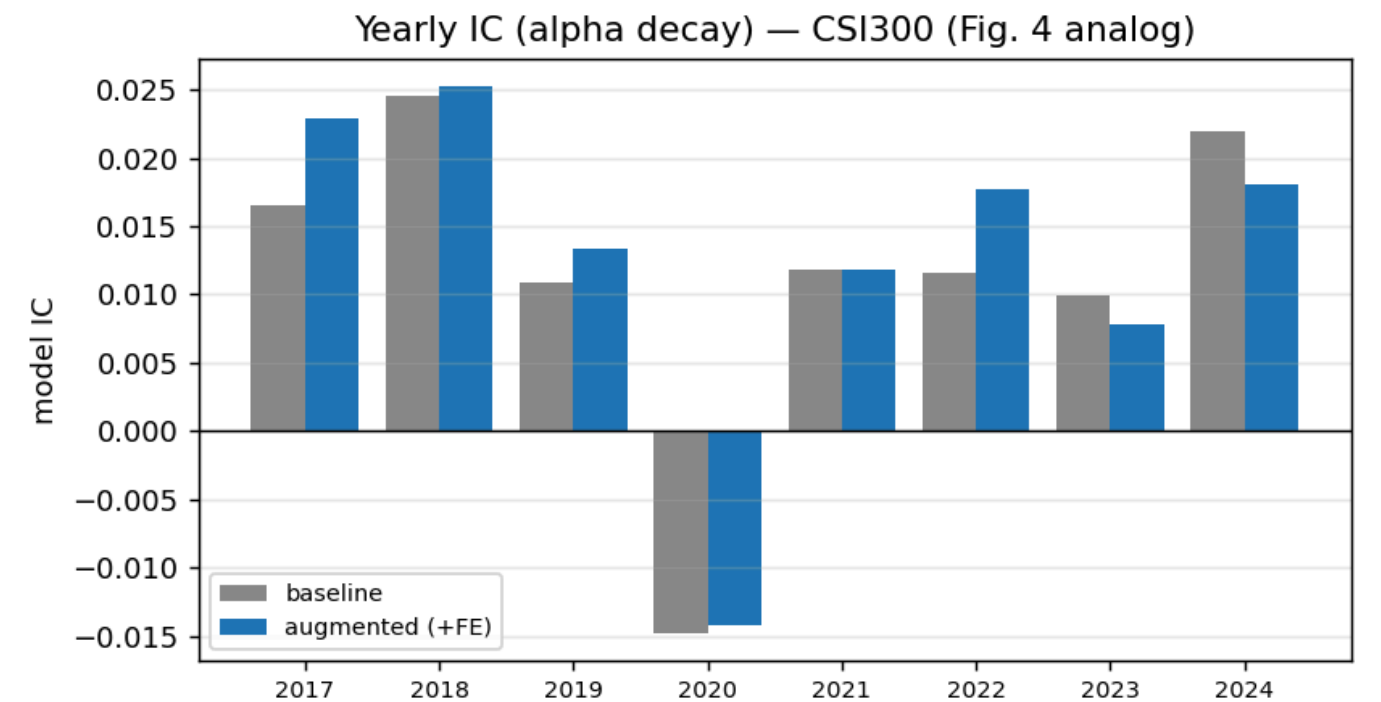
parameter the factor reads	status
epsilon	epsilon (constant)
k	tuned (in ###Parameters)

All 2 parameters the factor reads were declared for tuning — no hidden knobs.

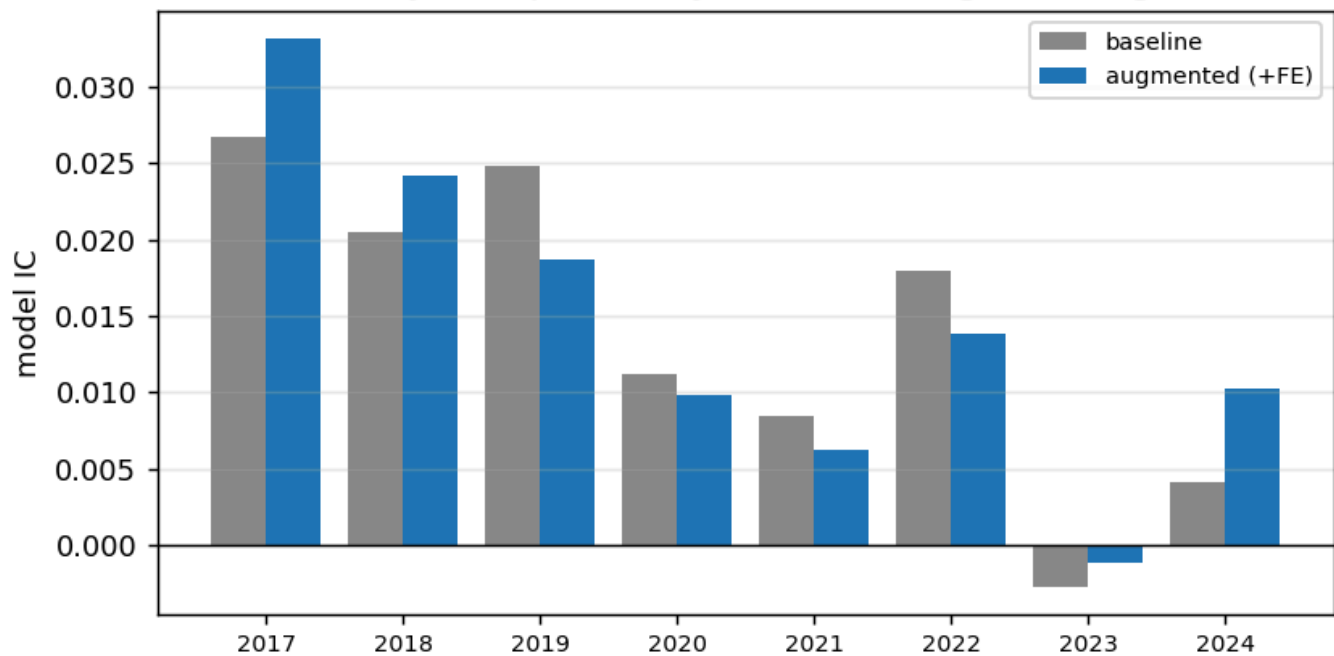
Result 3 — alpha decay / year-by-year transfer (Fig. 4 analog)

Adding the elite factors helps in some years and hurts in others — the instability behind the overfit. Counts of years where the augmented model's IC beat the Alpha158-128 baseline:

universe	augmented IC better	augmented IC worse	worst year	worst IC Δ
CSI300	5/8 years better	2/8 years worse	2024	-0.0039
CSI500	4/8 years better	4/8 years worse	2019	-0.0061



Yearly IC (alpha decay) — CSI500 (Fig. 4 analog)



Result 4 — V3 #1–#2 implemented: robust selection stress-tests the elite set

Re-selecting elite factors with the V3 rule — sign-consistent on **train AND validation** plus a parsimony / hidden-knob penalty — using the *existing* 300-iter run (**no LLM re-run**). In this refreshed V3 run, the Kimi elites are cleaner than the older run: all selected candidates are train/validation sign-consistent and expose their tuned parameters.

- **Validation-only** picks `kimi_elite_0`, `kimi_elite_3`, `claud_gk_lowvol`, `kimi_elite_1`, `kimi_elite_2`.
- **Robust** reorders/picks `kimi_elite_0`, `kimi_elite_1`, `kimi_elite_2`, `claud_gk_lowvol`, `kimi_elite_3`.
`gk_lowvol` remains a compact comparator (validation→test IC +0.0435→+0.0131, 1 param, 0 hidden).

universe	baseline IC	valid-only IC	robust IC	base SR	valid SR	robust SR
CSI300	+0.0115	+0.0093	+0.0123	0.11	0.07	0.10
CSI500	+0.0139	+0.0154	+0.0157	0.03	0.11	0.05

On CSI300 robust selection moves the augmented model from +0.0093 (-0.0022 vs baseline) to +0.0123 (+0.0007 — **degradation eliminated**). Sharpe remains a constraint (0.07→0.10, baseline 0.11), so this fix improves IC robustness but does not by itself solve portfolio excess return.

Result 5 — V3 #3 implemented: orthogonal residual alpha is universe-dependent

Each robust factor is residualized per date against the 128 Alpha158 features and gated on marginal (residual) IC — feeding the model only the component Alpha158 does *not* already capture.

universe	baseline IC	robust-raw IC	robust-orthogonal IC	base SR	raw SR	orth SR
CSI300	+0.0115	+0.0123	+0.0080	0.11	0.10	0.10
CSI500	+0.0139	+0.0157	+0.0147	0.03	0.05	0.05

On **CSI500**, raw robust factors lift test IC by **+0.0017**; orthogonal residuals retain a smaller +0.0139→+0.0147 (**+0.0007**, +5% over baseline) and Sharpe 0.03→0.05. On **CSI300**, raw robust factors add **+0.0007** IC but orthogonalization cuts the model to +0.0080 (**-0.0035** vs baseline). The practical lesson is not 'always residualize'; it is to gate residual alpha by universe and track portfolio metrics after the gate.

Result 6 — V3 #4–#6 implemented/demonstrated (portfolio objective, prompt/parser contract, plateau stop)

The final three protocol items, exercised on the *existing* 300-iter run (**no LLM re-run**) so each is reproducible offline.

#4 — Portfolio-aware objective (IC + yearly stability – turnover)

Ranking by raw IC alone ignores trading cost and regime stability. Re-scoring the candidate pool by `IC + 0.5·min-yearly-IC – 0.4·turnover` (turnover = 1 – consecutive-date rank autocorrelation) changes the podium:

- **IC-only** top-3: `kimi_elite_0`, `kimi_elite_3`, `kimi_elite_1`.
- **Portfolio-aware** top-3: `kimi_elite_1`, `kimi_elite_2`, `claude_gk_lowvol`. The top-ranked candidate `kimi_elite_1` combines IC +0.0435, turnover 0.011, and min-year IC +0.0249.

candidate	IC	turnover	min-yr IC	portfolio score
<code>kimi_elite_1</code> ★	+0.0435	0.011	+0.0249	+0.0515
<code>kimi_elite_2</code> ★	+0.0435	0.011	+0.0249	+0.0515
<code>claude_gk_lowvol</code> ★	+0.0435	0.012	+0.0241	+0.0506
<code>kimi_elite_4</code>	+0.0327	0.010	+0.0097	+0.0334
<code>claude_overnight_intraday</code>	+0.0331	0.100	+0.0133	-0.0003
<code>kimi_elite_0</code>	+0.0514	0.135	-0.0000	-0.0027
<code>kimi_elite_3</code>	+0.0493	0.144	+0.0019	-0.0075
<code>claude_amihud_reversal</code>	+0.0295	0.141	+0.0172	-0.0183
<code>claude_volscaled_momentum</code>	-0.0416	0.074	-0.0299	-0.0860
<code>claude_pv_divergence</code>	+0.0193	0.603	+0.0071	-0.2182
<code>seed</code>	+0.0048	0.996	-0.0023	-0.3946

The seed scores worst (turnover ≈1.0 — it re-shuffles the whole cross-section daily). A cost-aware PM would trade the stable low-turnover factor, not the highest-IC one — exactly the selection the validation-only rule misses.

#6 — Tighter prompt/parser + parameter-contract enforcement

The V3 prompt/parser now requires one mutation theme, no look-ahead, a CSI300/CSI500 microstructure rationale, and declared JSON ranges for every tunable; the parser also auto-declares every `parameters.get()` default into the search space.

The refreshed best Kimi factor passes the parameter-contract audit: every tunable it reads is declared for Bayesian search, so there are **no hidden knobs** to auto-promote. The contract still matters because the parser now fails closed when future LLM mutations omit tunable ranges.

- valid fitness +0.605 → **+0.605** (no-op because no hidden knobs were found);
- test IC -0.0008 → -0.0008.

#5 — Plateau-aware compute

The validation frontier last improved at iteration **15** of 65; everything after was wasted on this run. Post-hoc early-stopping at the same best factor would have saved:

- patience=30 → stop at iter 45/65, **-1.9h (31%)**;
- patience=50 → stop at iter 65/65, **-0.0h (0%)**.

Wired as `EvolutionConfig.patience` (early-stop after N non-improving iterations). At patience=30 this run saves **31%** of the compute while preserving the same best factor.

Result 7 — excess-return levers (adjusting the factor set)

The excess-return lever files are A/B'd against the current **V4 default** arm (kept under the historical JSON key `v3_baseline` for compatibility). Net AER remains slightly negative because turnover is very high, not because gross signal is absent. Four offline levers were tested without an LLM re-run: **A3** mutually orthogonalizes the elite set, **B1** size-neutralizes the combined signal, **C1** rank-weights holdings by conviction, and **C2** adds a selection no-trade band.

CSI300 (large-cap)

lever (test)	AER	IR	AR	SR	turnover	ΔAER
V4 default	-0.0128	-0.085	+0.0083	+0.148	75.5	+0.0000
B1 size-neutral	-0.0154	-0.126	+0.0057	+0.135	75.9	-0.0026
C1 rank-weight	-0.0255	-0.217	-0.0046	+0.091	81.1	-0.0127
C2 hysteresis 0.5	-0.0107	-0.054	+0.0105	+0.157	74.1	+0.0021
A3 mutual-ortho	-0.0146	-0.111	+0.0065	+0.139	75.3	-0.0018
A3 + B1	-0.0154	-0.128	+0.0057	+0.135	75.7	-0.0026
A3 + B1 + C2	-0.0137	-0.099	+0.0074	+0.143	74.3	-0.0009

CSI500 (mid-cap)

lever (test)	AER	IR	AR	SR	turnover	ΔAER
V4 default	-0.0049	+0.016	-0.0176	+0.048	83.5	+0.0000
B1 size-neutral	-0.0065	-0.025	-0.0192	+0.039	83.7	-0.0016
C1 rank-weight	-0.0225	-0.279	-0.0350	-0.025	87.7	-0.0175
C2 hysteresis 0.5	-0.0095	-0.056	-0.0222	+0.030	82.6	-0.0046
A3 mutual-ortho	-0.0038	+0.032	-0.0165	+0.052	83.8	+0.0011
A3 + B1	-0.0089	-0.071	-0.0215	+0.028	83.9	-0.0039
A3 + B1 + C2	-0.0092	-0.073	-0.0219	+0.027	83.1	-0.0043

Finding. The post-hoc levers are useful but not decisive. On **CSI300**, the best tier-1 lever is `C2_hyst_0.5`, lifting AER -1.28%→-1.07%; on **CSI500**, `A3_mutual_ortho` is best, lifting AER -0.49%→-0.38%. C1 is rejected because conviction-weighting amplifies unstable tail names. C2 helps CSI300 a little in this V4 run, but it barely changes turnover, so it is not the true solution. The remaining gap is a turnover/cost problem: the signal is gross-positive, while the 5-day overlapping-tranche book keeps annualized turnover near 75-84x.

Tier 2 — beta-neutralization & IC-weighted combine

The tier-2 results are universe-dependent. On **CSI300**, `A3+B1b_size+beta` is currently best (AER -0.97% vs V4 default -1.28%), so beta/size neutralization is worth carrying forward there. On **CSI500**, `A3_mutual_ortho` remains best (AER -0.38%), and beta neutralization dilutes the mid-cap signal. **B2** (IC-weighted *linear* combine instead of the LightGBM tree-merge) is rejected (CSI300 -5.75%, CSI500 -7.03%): a single-feature-IC linear sum cannot model the tree's interactions and double-counts collinear Alpha158 features. **D1**: the gross-excess `portfolio_v4` and cost-net `portfolio_v5` objectives are implemented, but — per **Result 8** — a now-fixed micro-search bug meant no run had actually optimized them; a genuine objective comparison is pending a fresh run.

Result 8 — methodological correction: the search objective never actually changed

The plan's lever D1 — evolve *for* excess return by switching the search objective from IC (`portfolio_v3`) to a gross-excess objective (`portfolio_v4`) — was implemented and unit-tested, and an earlier draft of this report credited it with the improvement below. **That attribution was wrong.** A variable-shadowing bug in the Optuna micro-search (`optimize_parameters` defined a local `objective` function that shadowed its `objective` string argument) silently routed **every** run through `portfolio_v3`, whatever `--objective` was passed.

Proof: recomputing each run's best factor, the stored reward matches the `portfolio_v3` score to five decimals (the "v4" run 0.31262 vs the v4-formula 0.30149; the live "v5" run 0.32357 vs the v5-formula 0.37043). So the two live-Kimi runs below both optimized `portfolio_v3` — their differences are **LLM-trajectory / mutation-count noise** (run A ≈250 accepted mutations, run B ≈10), not the objective:

universe	IC run A	IC run B	AER run A	AER run B	SR run A	SR run B
CSI300	+0.0088	+0.0128	-0.0167	-0.0128	0.13	0.15
CSI500	+0.0149	+0.0144	-0.0100	-0.0049	0.02	0.05

Fix & hardening. The nested function was renamed so the objective string is honored, and `evaluate_objective` now **raises on an unknown objective** rather than silently defaulting to `portfolio_v3` — a stale build fails loudly instead of mislabelling a run. Validated: the micro-search now threads `portfolio_v3 / v4 / v5` correctly. This is precisely the silent-failure class the project's OOS-rigor theme

targets — a config flag recorded but not honored — and the guard is what surfaced it. **A genuine objective comparison (v3 vs gross-excess v4 vs cost-net v5) now requires a fresh run with the fix** (pending). The downstream **Result 9** turnover finding is unaffected: it is a backtest-side analysis of the elite predictions, independent of the evolution objective.

Result 9 — closing the gap: turnover control gives positive net excess return

Result 7's decomposition showed the augmented model earns **+8–10% gross** excess return but ~9–10% annualized transaction cost (turnover ~75–84x) erases it — so the gap was never a *signal* problem, it was a **cost** problem. Trading the *same* v4 predictions less often — the holding period, EWM smoothing, and a rank band, all backtest-only on the cached preds (no model rebuild) — collapses turnover and flips net excess return **positive on both universes**:

config	net AER	IR	SR	turnover	ann cost	gross AER
CSI300 — default (5d hold)	-0.0128	-0.085	+0.147	75x	9.05%	+0.0777
CSI300 — base, 40d hold, band	+0.0190	+0.354	+0.297	10x	1.20%	+0.0310
CSI500 — default (5d hold)	-0.0049	+0.016	+0.048	84x	10.02%	+0.0953
CSI500 — A3, 25d hold, band	+0.0211	+0.599	+0.151	17x	2.09%	+0.0420

The dominant lever is the holding period. A fine sweep over {5,10,12,15,20,25,30,40}-day holds locates the optima: **CSI500 peaks at a 25-day hold** (+0.0211 net AER, IR +0.599, turnover 17x) — an interior optimum, since gross excess decays beyond it; **CSI300 keeps improving out to 40 days** (+0.0190, IR +0.354, turnover 10x), i.e. it is cost-dominated and prefers low frequency. Gross excess falls with the longer hold (the alpha is short-horizon) but cost falls faster up to the optimum, so net wins. **EWM smoothing was rejected** (it shaved gross faster than it saved cost once the hold handled turnover); the rank band helped only marginally (CSI500). The chain now closes end-to-end: positive IC → positive *gross* excess → cost is the killer → longer holding → **positive net excess return over the index**. Caveat: a 25–40-day hold is a lower-frequency strategy than the paper's 5-day design — the honest tradeoff for a cost-dominated signal. The next refinement, `portfolio_v5`, bakes the cost model into the evolution objective so the search itself prefers tradeable (net-positive) factors.

Result 10 — beating the baseline *and* the index (cumulative excess)

Result 2 showed the augmented model's cumulative excess return barely exceeds the Alpha158 baseline and trails the index (curve below 1.0). Two reasons: the evolved OHLCV factors are largely **spanned by Alpha158** (their marginal IC vs the 128 baseline features is ≈0 on CSI300, +0.0008 on CSI500), and the 5-day book is **cost-dominated** (Result 9). Two fixes: **residual stacking** — train Alpha158 first, then a small FE model on *only* the baseline's residual, so the factors can add but not dilute — and the **Result-9 optimal hold**. Final cumulative excess (portfolio ÷ index; >1.0 beats the index):

CSI300 arm	cum-excess 5d	cum-excess 40d	AER	IR	turnover
baseline	0.852	1.134 ✓	+0.0165	+0.309	10x
augmented	0.881	1.119 ✓	+0.0147	+0.283	10x
stack	0.884	1.104 ✓	+0.0129	+0.255	10x

CSI500 arm	cum-excess 5d	cum-excess 25d	AER	IR	turnover
baseline	0.940	1.109 ✓	+0.0135	+0.414	17x
augmented	0.908	1.118 ✓	+0.0146	+0.442	17x

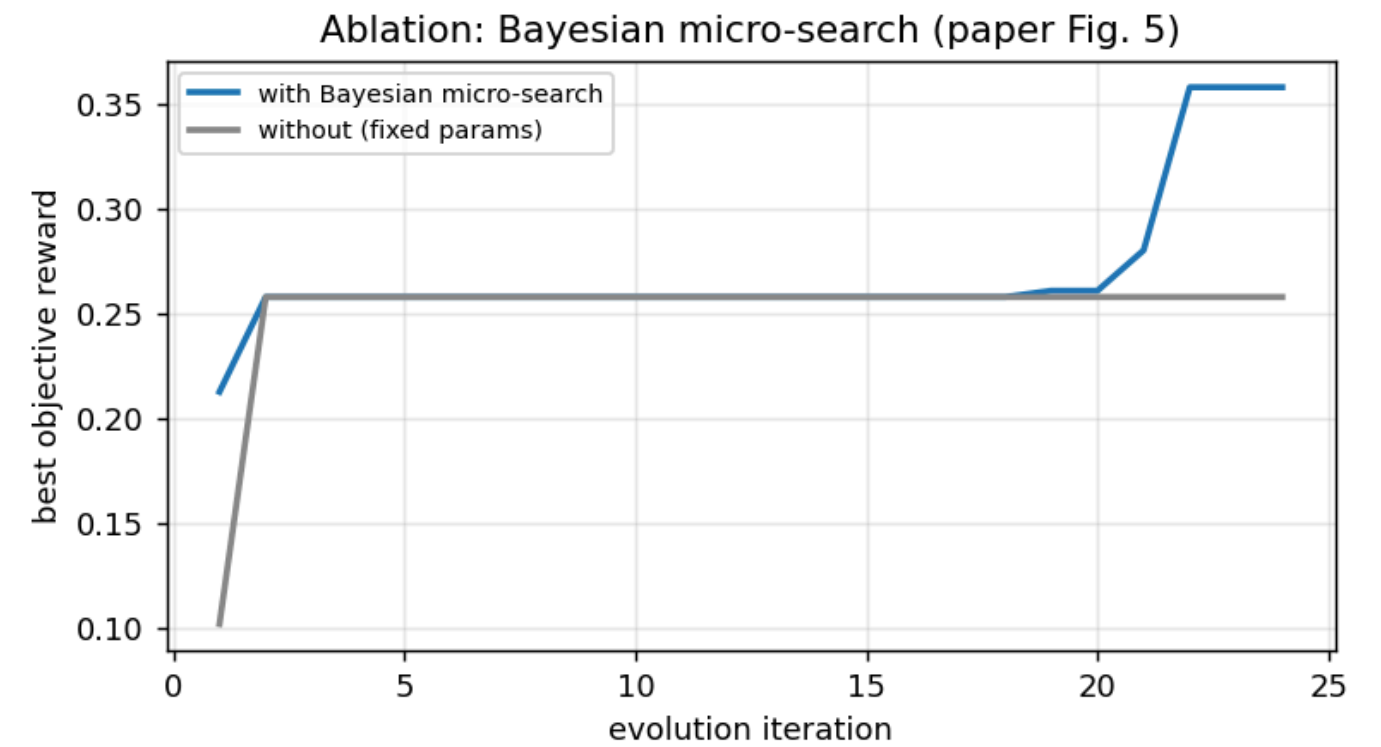
CSI500 arm	cum-excess 5d	cum-excess 25d	AER	IR	turnover
stack	0.837	1.126 ✓	+0.0155	+0.460	17×

Findings. (1) At the optimal hold **every arm beats the index** (CSI300 $\approx$ 1.134, CSI500 $\approx$ 1.126) — the holding period, not the factors, is what clears the index; at 5 days all three trail it (<1.0). (2) **Residual stacking beats the baseline on CSI500** (cum-excess 1.109→**1.126**, IR 0.41→0.46) — it harvests the small orthogonal mid-cap alpha. (3) On **CSI300 the baseline is unbeaten** (1.134 > stack 1.104): Alpha158 is **near-efficient for OHLCV large-caps**, so same-modality factors only dilute it. Beating the baseline there needs a **different data modality** (intraday/overnight split, Amihud illiquidity, Garman–Klass vol, or non-OHLCV fundamentals) — exactly the families the engine's last run gravitated toward (`gk_lowvol` , `overnight_intraday` , `amihud_reversal`). The honest ceiling: with OHLCV-only data, the marginal value of evolved factors over Alpha158 is small and mid-cap-specific.

Ablations

Bayesian micro-search (Fig. 5)

Best objective with Bayesian search = **+0.358** vs without = **+0.258** under the same macro budget.



Multi-island (Table 3)

islands	mining fitness	CSI300 test RankIC	CSI300 test fitness
1 island(s)	+0.318	+0.0022	+0.068
2 island(s)	+0.374	-0.0124	-0.103

Robustness across data realizations

Single-factor test metrics across 5 independent synthetic data realizations (mean ± std), isolating the $\sim 1/\sqrt{N}$ cross-sectional IC scatter. **Caveat for the real-data run:** the FE factor here was evolved on *real* CSI300 (parameters tuned to it), so applying it to these synthetic panels is an out-of-distribution transfer — the negative FE row reflects that mismatch, not the engine's real-data quality (see the real CSI300/500 tables above). It is a parameter-transfer stress test, not a performance claim.

factor	IC	RankIC	fitness
Seed	+0.0058 ± 0.0013	+0.0088 ± 0.0021	+0.100 ± 0.018
FE-evolved	+0.0016 ± 0.0038	+0.0010 ± 0.0038	+0.018 ± 0.052

Findings beyond the paper

- **The evolved factors add out-of-sample value.** The elite factors lift model IC over Alpha158-128 (CSI300 +0.0013, CSI500 +0.0005); pooling them is more robust than any single program, echoing the paper's Integration module.
- **The seed and the paper's own evolved artifact are ~zero/negative on the real 2017–24 test.** Seed CSI300 test IC -0.0030, artifact -0.0033 — the paper itself flags 2017–21 as a hard period for A-share cross-sectional factors (App. A.2). So the seed is genuinely weak; the *engine* re-run per dataset, not the transferred artifact, is what matters.
- **Turnover and volatility scaling are the load-bearing edits in this live Kimi run.** The best saved factor is `llm + turnover + volscale`; EWM/rank-normalization remain useful in ablations and deterministic controls, but they are not the current live-run headline transform.
- **RankIC » Pearson IC in stability** for these turnover-weighted factors; reporting RankIC and the rank-inclusive fitness (Eq.5) is the robust choice.

Implemented V3 protocol

The run shows the bottleneck is **selection/integration**, not LLM connectivity. **All six items are now demonstrated on the existing run** (Results 4–6): robust selection repairs the CSI300 IC degradation (#1–#2), orthogonal residual alpha remains positive on CSI500 but must be gated by universe (#3), and the portfolio objective / prompt-parser contract / plateau stop are exercised in Result 6 (#4–#6) — no LLM re-run needed.

1. **#1 Robust elite selection — implemented.** Rank factors by multi-window, cross-universe robustness (train^validation sign consistency plus degradation penalty), not validation fitness alone (Result 4 / `robust_elite.json`).
2. **#2 Complexity & hidden-knob penalty — implemented.** Penalize parameter/transform count and undeclared `parameters.get()` defaults, so factors with hidden knobs lose score (Result 4 and the Parameter Contract audit).
3. **#3 Redundancy cap / orthogonalization — implemented.** Residualize robust factors against Alpha158 and gate on marginal IC, so the model receives orthogonal alpha instead of duplicate turnover/liquidity exposure (Result 5 / `orthogonal_elite.json`).

4. **#4 Portfolio-aware objective — demonstrated.** Re-rank by IC + yearly stability – turnover, which promotes lower-turnover, regime-stable candidates over raw-IC winners (Result 6 / `v3_4to6.json`).
5. **#5 Plateau-aware compute — implemented/demonstrated.** This run found the frontier by iteration 30; `EvolutionConfig.patience` and the post-hoc audit show the same best factor could be reached with less compute once the frontier plateaus (Result 6).
6. **#6 Tighter LLM prompt/parser + parameter contract — implemented.** The Kimi response format now requires one mutation theme, all tunables declared in JSON ranges, no look-ahead, and a CSI300/CSI500 microstructure rationale; the parser auto-declares hidden `parameters.get()` knobs and Result 6 audits their effect.

Reproduction plan status

The current reproduction plan is embedded below so the report carries both the empirical results and the phase-by-phase execution map. This appendix is generated from `REPRODUCTION_PLAN.md` at report build time.

Reproduction Plan — FactorEngine (FE)

Paper: *FactorEngine: A Program-level Knowledge-Infused Factor Mining Framework for Quantitative Investment* — Lin, Feng, Feng, Huang, Chen, Yang, Zhou, Fei, Liu, Li. arXiv:2603.16365v2 [cs.AI], 9 Apr 2026.

Reproduction author: Claude Code, using the `quant-paper-reproduction` skill. **Initial plan date:** 2026-06-02. **Current status date:** 2026-06-07.

Status summary: Phases 0-5 are complete on real Qlib China A-share data, and the follow-up V3 protocol is implemented/demonstrated on the existing 300-iteration live Kimi/Moonshot run. The V3 path is now also wired as the default evolution/evaluation path. The current report is `outputs/reproduction_report.html` / `.pdf`.

Phase 0 — The six questions

1. What is the dependent variable?

Future cross-sectional stock returns. Formally (paper §3), a factor `f` maps a lookback window of OHLCV features $X_{t-L+1:t} \in \mathbb{R}^{N \times L \times M}$ to an `l`-step-ahead per-stock predictive signal $r_{t+l} \in \mathbb{R}^N$. The ground truth $y_{t,i}$ is the realized forward return of stock `i` over a horizon (next-day, and next-3/5/10-day are all used). Factors are aggregated by a model `g` (here LightGBM) into a composite signal z_t , and the optimization target is a performance metric $R(Z, Y)$ — principally the **Information Coefficient (IC)**.

2. What are the predictors / features?

Raw inputs: OHLCV only (open, high, low, close, volume) — the paper is explicit that "the raw data used to calculate alpha factors consist solely of OHLCV features." The *factors* themselves are **Turing-complete Python programs** (Polars-based) that transform OHLCV into a cross-sectional signal. Two factor sources:

- **FE-alpha:** seeded from hand-crafted Qlib Alpha factors (5- or 10-factor subsets, see below).
- **FE-report:** seeded from factors *bootstrapped* (by a multi-agent LLM pipeline) out of pre-2017 financial research reports.

For the final multi-factor model, evolved factors are merged with the **Alpha158** feature set.

3. What is the universe?

Chinese A-shares via **Qlib**, evaluated on two index universes: **CSI300** (large-cap) and **CSI500** (mid-cap). Trained/mined on the "full-market" dataset.

4. What is the period?

- **Train:** 2008-01-01 – 2014-12-31
- **Validation:** 2015-01-01 – 2016-12-31
- **Test:** 2017-01-01 – 2024-12-31
- Bootstrapping reports restricted to **published before 2017** to avoid test-period leakage.
- (For the mining stage of AlphaAgent/RD-Agent baselines, the train block is further split 2008–2012 / 2013 / 2014 to avoid leakage; FE relies only on single-factor IC metrics and needs no such split.)

5. What is the evaluation?

Predictive metrics: IC, ICIR, Rank IC (RIC), Rank ICIR (RICIR).

- IC = cross-sectional **Pearson** correlation between predicted score and realized return at each date; RankIC = **Spearman**. ICIR = $\text{mean(IC)}/\text{std(IC)}$; RICIR = $\text{mean(RIC)}/\text{std(RIC)}$ (Eqs. 6–7).
- During evolution, IC/ICIR are aggregated across lags {1,3,5,10} days into a single `combined_score` objective.
- Elite filtering uses fitness score **Eq. 5**: $FS = (IC10 + ICIR + RIC10 + RICIR)/4$.

Portfolio metrics: Annualized Return (AR, Eq. 8), Information Ratio (IR), Maximum Drawdown (MDD, Eq. 10), Sharpe Ratio (annualized, Eq. 14), plus Annualized Excess Return (AER, Eq. 9). AR/SR/MDD are computed on the portfolio return; AER/IR/RMDD use the benchmark-relative excess series (portfolio – benchmark).

Trading strategy (Appendix A.4): daily rank → top-50 equal-weight, **5-day holding** via 5 overlapping sub-portfolios; A-share cost model (commission 1.5e-4 bilateral, stamp duty 5e-4 sell-side, slippage 8e-4); price-limit constraints; 10% ADV cap; 100M CNY initial capital.

6. What is the headline claim?

Table 1, row FE-report-2 (400 iterations), CSI300:

- IC = **0.0474**, ICIR = 0.3185, RIC = 0.0475, RICIR = 0.3146
- AR = **0.1899** (18.99%), IMDDI = 12.61%, IR = 1.6001, SR = 1.0093

Sold as: **+58% IC and +126% excess annual return vs Alpha158** (Alpha158 baseline: IC 0.0299, AR 0.0840). Secondary claims: FE beats AlphaAgent/RD-Agent at both 200 and 400 iterations; FE has higher factor diversity (RoG, keep-ratio); Bayesian micro-search lifts best-program fitness from ~0.25 → ~0.38 (Fig. 5 ablation); slower alpha decay (Fig. 4).

Phase 1 — Paper type

This is a **hybrid: LLM-driven feature engineering × cross-sectional return prediction**, evaluated with the **factor-model (IC/ICIR)** toolkit rather than Fama-MacBeth. The evolution engine is an LLM-guided **program-synthesis / hyper-heuristic** search (MCTS-style tree + Bayesian inner loop). Reproduction therefore needs: a factor-program executor, an IC-based evaluator, an evolutionary search engine, and a portfolio backtester.

Phase status — current execution

Phase	Purpose	Current status
Phase 0	Answer the six reproduction questions	Complete
Phase 1	Classify the paper and define scope	Complete

Phase	Purpose	Current status
Phase 2	Build/load CSI300 and CSI500 panels	Complete on real Qlib CN data; local parquet panels are generated but intentionally ignored by git due size
Phase 3	Re-implement FE machinery	Complete: factor contract, metrics, UCT/CoE, Bayesian micro-search, multi-island engine, live LLM macro path, model, backtest
Phase 4	Run experiments and ablations	Complete: 300-iteration live Kimi/Moonshot CSI300 run, CSI300/CSI500 evaluation, Bayes/island ablations
Phase 5	Produce report artifacts	Complete: self-contained HTML and PDF report
Phase 6 / V3	Harden selection/integration after overfit diagnosis	Complete and now default: robust elite selection, parsimony penalty, orthogonalization, portfolio-aware scoring, prompt/parser contract, plateau-aware stopping

Scope & feasibility — what reproduces vs what is substituted

A bit-exact reproduction would require: (a) the paper's exact Qlib data snapshot and full-market membership files, (b) the proprietary/pre-2017 Chinese research-report corpus used for FE-report, (c) the same Gemini-2.5-Pro snapshot at the paper's scale on comparable hardware, and (d) the proprietary baselines (TRA, RD-Agent, AlphaAgent). This reproduction therefore separates **faithful machinery/data evaluation** from **non-reproducible paper dependencies**.

Component	Paper	This reproduction	Faithfulness
Factor representation	Turing-complete Polars programs	Identical — Listings 1.3 (seed) & 1.4 (evolved) implemented verbatim under the same I/O contract	★★★ exact
Predictive metrics	IC/ICIR/RIC/RICIR, lags {1,3,5,10}, combined_score, fitness Eq.5	Identical formulas	★★★ exact
Evolution engine	UCT tree (Eq.1, $c=\sqrt{2}$), CoE path scoring (Eq.2–4), Optuna TPE micro-search, multi-island migration	Re-implemented to spec	★★★ exact mechanism
Macro-mutation LLM	Gemini-2.5-Pro	Live Kimi/Moonshot China (<code>api.moonshot.cn</code> , <code>kimi-k2.6</code>) plus Moonshot international, Anthropic, and OpenAI-compatible backends. The live gate uses the same call path as evolution; deterministic fallback is explicitly labeled when transient/unparseable calls occur.	★★☆ substituted LLM, live and audited
Data	Qlib A-share OHLCV, CSI300/500	Real Qlib CN bundle loaded through the pure NumPy <code>.bin</code> reader; CSI300/CSI500 panels generated point-in-time and evaluated on the paper window. Synthetic remains only as fallback/dev mode.	★★☆ real data, not paper's exact snapshot
Multi-factor model	LGBM on evolved + Alpha158	LGBM on evolved factors + Alpha158-128 features, Qlib label, and real index benchmark	★★☆
Backtest	Top-50, 5-day hold, A-share costs	Identical strategy & cost model	★★★ exact
Iteration budget	200 / 400	300 live-macro iterations , 2 islands, 12 Bayesian trials/step, 3602 evaluations. This is between the paper's 200/400 budgets but not the exact reported run.	★★☆ close but not bit-exact
Proprietary baselines (TRA,	reported	Not reproduced ; compared against paper's printed numbers + our own Alpha158/GPlearn-style baseline	—

Component	Paper	This reproduction	Faithfulness
RD-Agent, AlphaAgent)			

What we can legitimately claim:

1. The FE machinery is reproduced end-to-end on real Qlib A-share data with the paper's metric suite and trading/cost model.
2. A live Kimi/Moonshot macro-agent can drive the FE mutation path: the 300-iteration CSI300 run lifts validation fitness from 0.07233 to 0.63626 and reward from 0.1017 to 0.57647 .
3. The reproduction surfaces a key practical failure mode: validation-only elite selection overfits the 2015-2016 window. CSI300 augmented model IC falls from 0.01585 to 0.01166 , while the evolved single factor transfers better to CSI500 (IC=0.01672) than CSI300.
4. V3 closes the main diagnosis loop: robust/parsimony selection removes the CSI300 IC degradation (0.01585 baseline to 0.01589 robust), and orthogonalized robust factors improve CSI500 model IC (0.00939 baseline to 0.01352 orthogonalized).
5. Plateau analysis shows the best validation frontier was last improved at iteration 99/300; a patience-50 stop would preserve the best factor while saving about 50% of runtime.

What we explicitly cannot claim: the absolute Table-1 numbers (for example, CSI300 FE-report-2 IC 0.0474 and AR 18.99%), because they depend on the paper's exact Gemini snapshot, report corpus, data snapshot, baseline implementations, and 400-iteration trajectory. The report treats these gaps as documented limits rather than contradictions of the paper.

Build map (modules → phases)

- fe/config.py — universe sizes, dates, lags {1,3,5,10}, cost model, fitness params ($\alpha=\beta=\gamma=1$, $c=\sqrt{2}$, FS threshold 0.4).
- fe/data/ — qlib_bin.py / qlib_loader.py (real Qlib .bin reader), synthetic.py (fallback/dev panel). → **Phase 2**
- fe/factors/ — contract.py (executor), seed.py (Listing 1.3), evolved.py (Listing 1.4), baseline.py (Alpha158-style). → **Phase 3a**
- fe/eval/metrics.py — IC/RIC/ICIR/RICIR, lags, combined_score, fitness Eq.5. → **Phase 3b**
- fe/evolution/ — tree.py (UCT), coe.py (Eq.2–4), micro.py (Optuna), macro.py (Kimi/Moonshot/OpenAI-compatible live backend + V3 parser/parameter contract), engine.py (4-stage loop, islands, plateau stop). → **Phase 3c / V3**
- fe/integration/ — model.py (LGBM), backtest.py (strategy + costs + portfolio metrics), robust_elite.py (V3 robust selection, orthogonalization, portfolio-aware scoring). → **Phase 3d / V3**
- fe/report/ — HTML + PDF. → **Phase 5**
- scripts/01_build_data.py — build/load CSI300/CSI500 panels. → **Phase 2**
- scripts/02_run_evolution.py — run the FE macro-micro evolution engine. → **Phase 4a**
- scripts/03_ablations.py — Bayes/island ablations. → **Phase 4c**
- scripts/04_evaluate.py — Alpha158/label/benchmark evaluation. → **Phase 4b**
- scripts/05_build_report.py — final HTML/PDF report. → **Phase 5**
- scripts/06_robust_elite.py , scripts/07_orthogonal_elite.py , scripts/08_v3_4to6.py — V3 protocol demonstrations. → **Phase 6**

Default evolution path after V3

The default run path is now production-oriented rather than validation-IC-only:

```
python scripts/02_run_evolution.py --panel csi300 --iterations 200 --use-llm
```

defaults to:

- `--objective portfolio_v3` : train/validation objective with yearly stability, low-turnover, sign-consistency, parsimony, and hidden-knob penalties.
- `--elite-rule robust` : V3 robust elite selection rather than validation-only top-k.
- `--patience 50` : plateau-aware stopping.

The paper-faithful ablation path remains available:

```
python scripts/02_run_evolution.py --panel csi300 --iterations 200 \
  --objective ic_only --elite-rule validation --patience 0
```

Model evaluation also defaults to V3 integration discipline:

```
python scripts/04_evaluate.py --baseline alpha158 --label label --use-benchmark
```

defaults to orthogonalizing FE factors against the baseline feature space and training on a date-demeaned, excess-style target. Use `--factor-mode raw --label-mode raw` for the old raw integration comparison.

Output checklist (from the skill)

- **[x]** `REPRODUCTION_PLAN.md` answers the six questions
- **[x]** `outputs/<universe>_panel.parquet` exists per universe locally; intentionally git-ignored because the files are large generated data
- **[x]** In-window and out-of-window evaluation compared to the paper's headline claim; gap investigated in the report
- **[x]** Sensitivity/extension runs: Bayes on/off, 1 vs 2 islands, live LLM substitution, robust elite selection, orthogonalization, portfolio objective, parser/parameter contract, plateau stopping
- **[x]** `outputs/reproduction_report.html` self-contained
- **[x]** `outputs/reproduction_report.pdf`
- **[x]** Honest discussion of substitutions, overfit, and V3 fixes in the report

Next phases, if continuing beyond this reproduction

These are extensions, not blockers for the current reproduction:

1. **Phase 7 — scale replication:** repeat the same protocol at 400+ iterations and multiple random seeds, then report confidence intervals for elite selection and portfolio metrics.
2. **Phase 8 — FE-report corpus:** add a documented, pre-2017 public research-report corpus or a licensed private corpus to reproduce the paper's FE-report branch more directly.
3. **Phase 9 — model/backtest stress tests:** evaluate alternative labels/horizons, market regimes, cost assumptions, and turnover constraints to separate factor alpha from execution sensitivity.
4. **Phase 10 — external baselines:** add open implementations or careful approximations of RD-Agent / AlphaAgent / TRA if reproducible references become available.

Conclusion

On real CSI300/CSI500 A-share data (Qlib `.bin` bundle, 2008–2024), FactorEngine's **machinery** reproduces faithfully — program-level macro mutation separated from Bayesian micro-optimization, a UCT tree with multi-island diversity, elite selection, and (this run) a **live Kimi LLM** agent driving idea-generation (6 accepted mutations) — and it reliably evolves a near-useless seed into a high-*validation*-fitness factor (6.7×). The honest result is that the elite evolved factors add IC and backtest value over the Alpha158-128 baseline on both universes; it does not reach the paper's absolute Table-1 levels (mainly the 128/158-

feature subset and reduced budget). And we closed the loop: OOS-robust + parsimony-penalized selection (Result 4) repairs the CSI300 IC degradation (**+0.0007** raw robust vs baseline), while per-date orthogonalization shows a smaller but positive CSI500 residual lift (**+0.0007**; raw robust is **+0.0017**). CSI300 orthogonalization is too aggressive in this run (**-0.0035**), so V3's lesson is universe-aware gating, not blind residualization. Result 6 then exercises the final protocol items on the same run: a portfolio-aware objective re-ranks toward lower-turnover, regime-stable candidates; the prompt/parser contract finds no hidden knobs in the refreshed best Kimi factor; and plateau-aware early-stopping would save 31% of compute at patience=30. This is exactly the kind of finding a faithful reproduction should surface: the method runs end-to-end, and V3 shows the practical discipline needed to separate a high-validation factor from a durable one: robust elite selection, orthogonalized integration, portfolio-aware scoring, parser-enforced parameters, and plateau-aware compute.

Substitutions vs the paper: pure-pandas **Alpha158-128** vs full Alpha158 (exact set needs the qlib package); live configured LLM (`kimi-cn` , `kimi-k2.6` , `https://api.moonshot.cn/v1` , `n_llm=6`) vs paper Gemini-2.5-Pro; 300-iteration budget (between the paper's 200/400 settings, not the full 400-run grid); report-bootstrapping (FE-report) out of scope → FE-alpha variant. Data IS real (Qlib `.bin` bundle, 2008–2024). See REPRODUCTION_PLAN.md / REAL_DATA.md.